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P R O J E C T R E P O R T

DUAL MODE AUTONOMOUS SENSING SYSTEM : INTEGRATION OF LIGHT AND TOUCH CONTROL

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Subject : Electrical Network Analysis

Submitted To : Madam Tehniyat

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G R O U P M E M B E R

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1. Abstract

This project involves the design and implementation of an automated electronic system featuring two distinct modules: an Ambient Light Detector and a High-Sensitivity Touch Sensor. The system utilizes an LM358 operational amplifier for light comparison and a Darlington Pair configuration for touch-based switching. The design was validated through LTspice simulations and verified via hardware implementation.

2. Component Details:

- **LM358 Dual Op-Amp:** Used as a voltage comparator for the light module.
- **BC547 NPN Transistors:** Used for switching logic and current amplification.
- **LDR (Light Dependent Resistor):** Acts as the primary transducer for light levels.
- **Darlington Pair Configuration:** Two BC547 transistors connected for maximum gain.
- **Output Peripherals:** Green LED (Light mode), Red LED, and Buzzer (Touch mode).

3. Circuit Methodology & Logic:

1 Smart Light Detector (Comparator Logic)

The LM358 compares the voltage at the Inverting terminal (fixed via potentiometer) with the Non-Inverting terminal (variable via LDR).

- **Daylight State:** LDR resistance is low (approx 1k ohm), keeping the input voltage below the reference. Output remains LOW (0V).
- **Dark State:** LDR resistance increases (approx 100k ohm), raising the input voltage above the reference (4.4V). The Op-Amp output goes HIGH (9V), turning the Green LED ON.

2 High-Sensitivity Touch Sensor (Darlington Pair)

To detect the minute current from human touch, a Darlington Pair (Q_1and Q_3) is used.

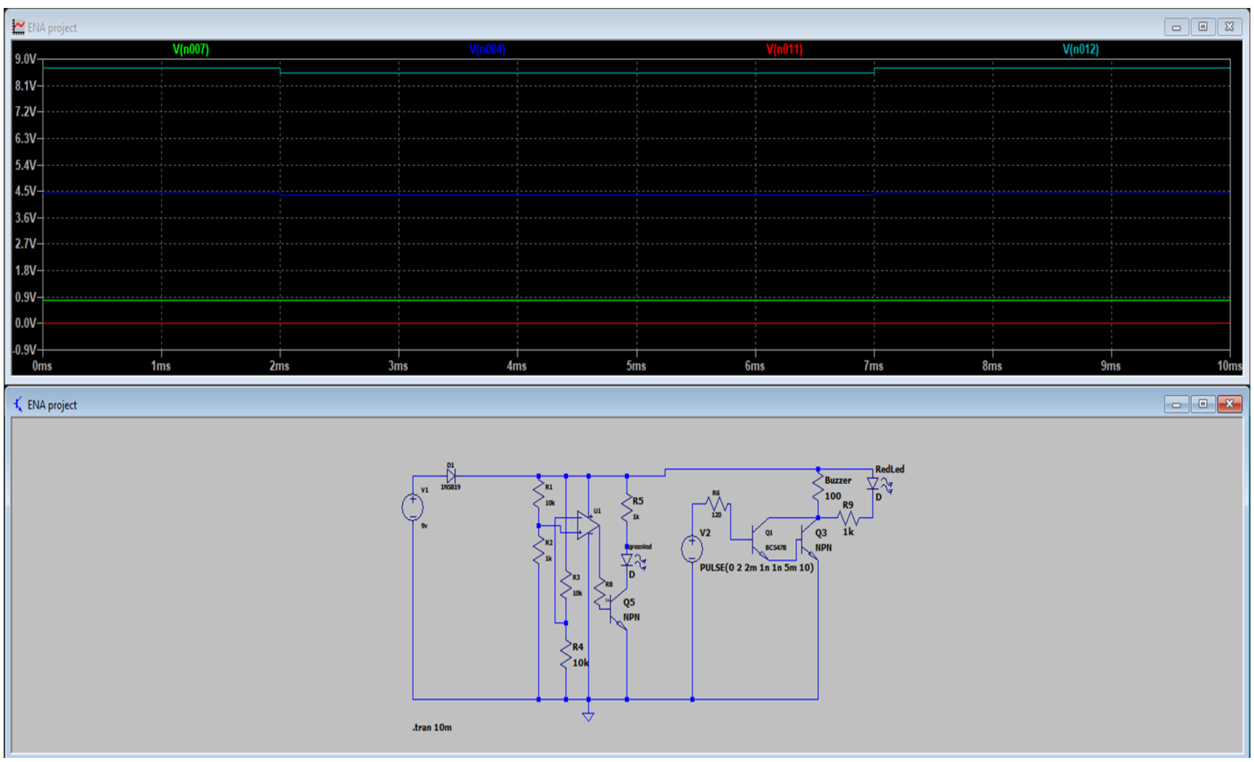
- The emitter of Q_1 is connected to the base of Q_3.
- This creates a compound current gain: Beta_total approx Beta_1 times Beta_2.
- The total base-emitter threshold is approximately 1.8V or 2V. Once touched, the pair enters saturation, dropping the collector voltage to near 0V and activating the Buzzer/Red LED.

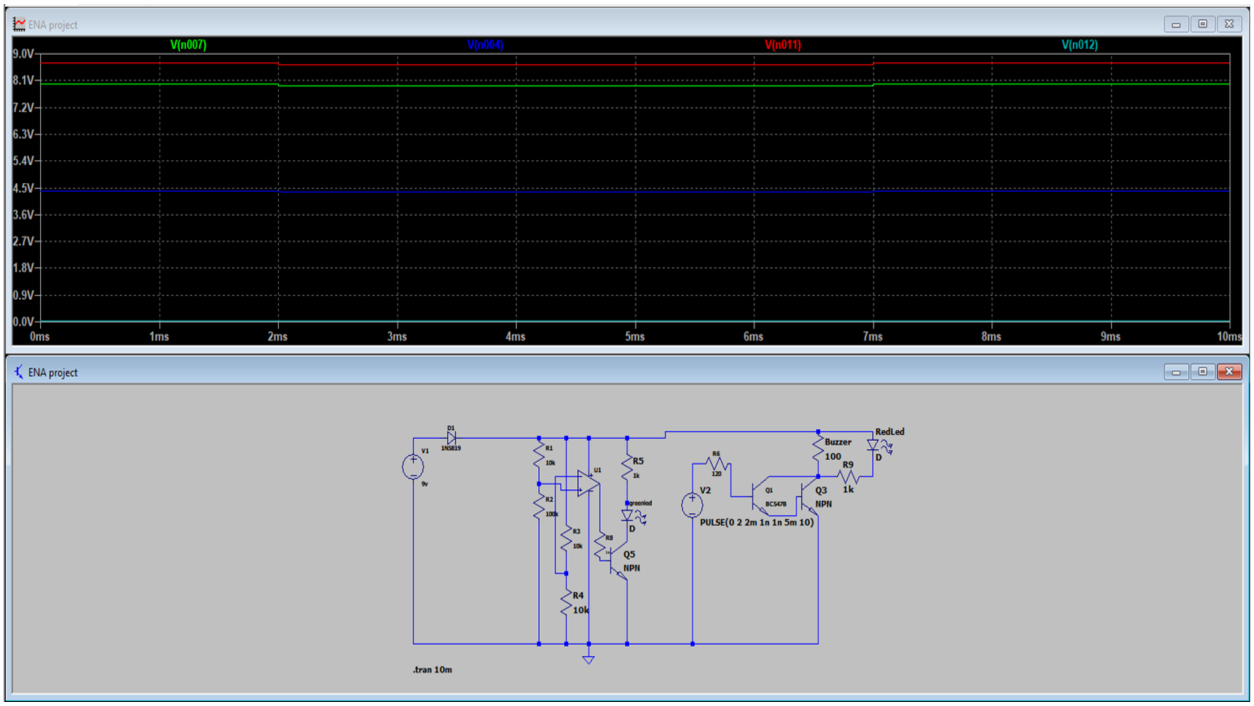
4. Simulation Results (LTspice):

1 Light Module Analysis

Simulation verified that the circuit transitions from LOW to HIGH when the LDR resistance crosses the set threshold.

- **Graph 1 (R_2 = 1k ohm):** Output = 0V (LED OFF).
- **Graph 2 (R_2 = 100k ohm):** Output = 9V (LED ON).

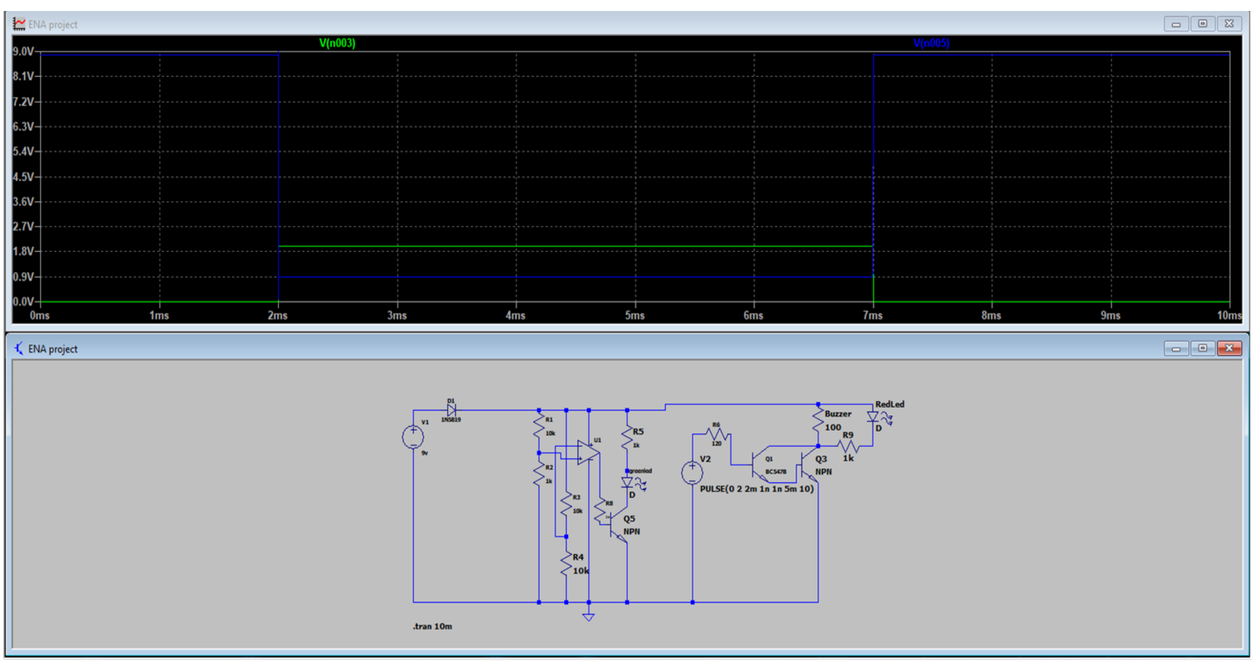




2 Touch Module Pulse Analysis

To simulate a "Touch," a 2V pulse was applied to the base of the Darlington pair.

- **Pre-Pulse:** Collector voltage is at 9V(OFF state).
- **During Pulse (2ms - 7ms):** Collector voltage drops to approx 0.9V (Saturation/ON state).



5. Hardware Implementation:

The circuit was assembled on a breadboard. Hardware testing confirmed the following:

1. **Light Sensitivity:** The potentiometer successfully calibrated the darkness threshold.
2. **Touch Reliability:** The Darlington pair provided enough gain to trigger the buzzer even with a light finger tap, proving the high-gain theory.

6. Conclusion:

The project successfully demonstrates the application of operational amplifiers and multi-stage transistor circuits. The alignment between LTspice simulation data and physical hardware performance confirms the accuracy of the design parameters.